=== UNIT 1: GRAMMATICAL FOUNDATIONS & PARTS OF SPEECH ===

>> 1.1 Structural Taxonomy of English Parts of Speech

Professional scientific communication demands rigorous adherence to grammatical syntax, clarity, and precision.
* Nouns and Nominalization: Proper, Common, Collective, Abstract, and Material nouns.
- Countable vs. Uncountable Nouns: Countable nouns (cell, enzyme, molecule) take plural forms and indefinite articles. Uncountable nouns (biomass, DNA, water, equipment, information) take singular verbs and cannot take indefinite articles or plural suffixes (*equipments is strictly ungrammatical).
* Pronouns: Subjective (I, he, they), Objective (me, him, them), Possessive (my, his, their), Demonstrative (this, that, these, those), Relative (who, which, that, whose).
- Relative Pronoun Distinction: 'Which' introduces non-restrictive (parenthetical) clauses preceded by a comma; 'That' introduces restrictive (essential) clauses without commas.
* Verbs: Finite (carries tense, number, person) vs. Non-Finite (Gerunds [-ing functioning as noun], Infinitives [to + verb], Participles).
- Transitive verbs require a direct object (e.g., 'The scientist synthesized the peptide'); Intransitive verbs do not take direct objects (e.g., 'The precipitate settled').
* Adjectives and Degree of Comparison: Positive, Comparative (-er / more), and Superlative (-est / most). Avoid redundant double comparatives (*more higher).
* Adverbs: Modify verbs, adjectives, or other adverbs. Categorized by Manner (rapidly), Frequency (regularly), Degree (substantially), and Time (subsequently).
* Prepositions: Spatial (in, on, at, under, across), Temporal (at [exact time], on [days/dates], in [months/years/durations]), and Directional (into, onto, toward).
- Between vs. Among: 'Between' is used for distinct individual entities (typically two); 'Among' is used for collective groups of three or more.
* Conjunctions: Coordinating (FANBOYS: For, And, Nor, But, Or, Yet, So), Subordinating (although, because, since, whereas, while), Correlative (either...or, neither...nor, not only...but also).

[EXERCISE 1: Error Identification and Correction in Technical Literature]
Problem: Identify and correct grammatical errors in the following four academic biotechnology sentences, providing rigorous linguistic justifications.
Sentence 1: "The laboratory purchased several new equipments and hardwares for the molecular biology department."
- Error: 'equipments' and 'hardwares' are mass/uncountable nouns in standard English; they do not possess plural -s forms.
- Correction: "The laboratory purchased several new pieces of equipment and hardware for the molecular biology department."
Sentence 2: "The concentration of the solute in the beaker is more higher than the surrounding matrix."

- Error: 'more higher' is a redundant double comparative. The comparative form of high is 'higher'.
- Correction: "The concentration of the solute in the beaker is significantly higher than that of the surrounding matrix."

Sentence 3: "Each of the researchers have submitted their respective progress reports to the principal investigator."
- Error: 'Each' is an indefinite pronoun that is grammatically singular and takes a singular verb ('has').
- Correction: "Each of the researchers has submitted his or her progress report to the principal investigator."

Sentence 4: "The enzyme solution was divided equally between the four test tubes."
- Error: 'Between' applies strictly to two distinct items; for four test tubes, 'among' is required.
- Correction: "The enzyme solution was divided equally among the four test tubes."

[EXERCISE 2: Prepositional Accuracy in Experimental Protocols]
Problem: Fill in the correct prepositions (at, on, in, into, between, among, of, with) in the following scientific laboratory protocol:
Protocol Passage:
"The experiment commenced [1] Monday morning [2] 09:00 AM. 5 mL of cell suspension was transferred [3] a sterile centrifugation tube. The mixture was incubated [4] a constant temperature [5] 37 deg C [6] a water bath for 48 hours. Variations in biomass yield were observed [7] the three replicates."
Solution and Grammatical Rule:
[1] on: 'on' is mandatory before days of the week (on Monday).
[2] at: 'at' indicates specific points in time (at 09:00 AM).
[3] into: 'into' indicates movement toward the interior of a container (transferred into).
[4] at: 'at' is used for specific physical measurements/temperatures (at a constant temperature).
[5] of: 'of' expresses quantity or specification (of 37 deg C).
[6] in: 'in' denotes enclosure within an apparatus (in a water bath).
[7] among: 'among' is required when distributing across three or more items (among the three replicates).

[EXERCISE 3: Correlative Conjunction Parallelism in Technical Writing]
Problem: Re-write the following sentences to enforce strict syntactic parallelism across correlative conjunction pairs (not only...but also, either...or, neither...nor).
Sentence 1: "The newly designed bioreactor not only improved oxygen transfer rate, but it also was reducing operational energy costs."
- Error: Asymmetric verb tenses. 'improved' (simple past) vs. 'was reducing' (past continuous).
- Parallel Revision: "The newly designed bioreactor not only improved oxygen transfer rate, but also reduced operational energy costs."

Sentence 2: "The technician must either recalibrate the spectrophotometer or the results will be invalid."

- Error: Faulty placement of 'either'. Preceding a verb in clause 1, but following an entire clause in part 2.
- Parallel Revision: "Either the technician must recalibrate the spectrophotometer, or the experimental results will be deemed invalid."
Sentence 3: "The compound exhibited neither antimicrobial activity against E. coli nor did it inhibit S. aureus."
- Parallel Revision: "The compound exhibited neither antimicrobial activity against E. coli nor inhibitory effects against S. aureus."

[EXERCISE 4: Relative Pronoun Disambiguation: Restrictive vs. Non-Restrictive]
Problem: Contrast the meaning and punctuation of the relative clauses in the following pair of sentences:
Sentence A: "The centrifuges that were manufactured in Germany operated without vibration."
Sentence B: "The centrifuges, which were manufactured in Germany, operated without vibration."
Analysis:
- Sentence A uses 'that' without commas (Restrictive Clause). The clause defines WHICH specific centrifuges operated without vibration. Implication: The laboratory owns multiple centrifuges; ONLY the subset manufactured in Germany operated smoothly, whereas centrifuges from other countries did not.
- Sentence B uses 'which' enclosed in commas (Non-Restrictive / Parenthetical Clause). The geographic origin is incidental background information. Implication: ALL the laboratory's centrifuges operated without vibration; additionally, as a matter of fact, they happen to have been manufactured in Germany.

>> 1.2 University Review & Examination Questions

* Q1: Define Countable and Uncountable nouns and list five common uncountable nouns in scientific writing that are frequently misused with plural suffixes.
Answer: Countable nouns refer to discrete, individual entities that can be counted with cardinal numbers, take indefinite articles (a/an) in the singular, and form regular or irregular plurals (e.g., pipette/pipettes, organism/organisms). Uncountable (mass) nouns refer to undifferentiated materials, concepts, or collective substances that cannot be divided into separate units; they take singular verbs and cannot take indefinite articles or plural suffixes. Five frequently misused uncountable nouns in technical prose: (1) Equipment (*equipments is incorrect; use 'pieces of equipment'), (2) Information (*informations is incorrect; use 'pieces of data/information'), (3) Apparatus (*apparatuses is incorrect when used as a mass noun), (4) Research (*researches as a noun is incorrect; use 'research studies'), (5) Software (*softwares is incorrect; use 'software programs').
* Q2: Explain the syntactic rule of parallelism in technical English with illustrative examples involving coordinating and correlative conjunctions.
Answer: Parallelism requires that two or more elements joined by coordinating conjunctions (and, but, or) or correlative conjunctions (either...or, neither...nor, not

only...but also) must share identical grammatical form (e.g., noun matched with noun, gerund with gerund, infinitive with infinitive, clause with clause). Violation example: 'The engineer is responsible for designing the vessel, fabrication, and to test the pressure.' Here, a gerund (designing), a noun (fabrication), and an infinitive (to test) are mixed. Parallel correction: 'The engineer is responsible for designing the vessel, fabricating the structure, and testing the pressure' (all three elements are parallel gerunds).

>> 1.3 Key Takeaways and Revision Points

* Mass nouns (equipment, information, biomass) are singular and never take plural -s endings.
* Restrictive clauses use 'that' without commas; non-restrictive clauses use 'which' enclosed in commas.
* Prepositions demand precision: 'at' for specific time/points, 'on' for days, 'in' for years/enclosures, 'between' for 2, 'among' for >=3.
* Correlative conjunctions (not only...but also) require identical grammatical structures on both sides.